Constraint-Driven Geometric Substrate Selection in Self-Referential Fields:

From Renormalization Collapse to Multi-Scale Structure

A Correction and Extension of the CQFT ϕ-Fixed-Point Hypothesis

Daniel Solis

DUBITO Inc. — Dubito Ergo AGI Safety Project

Czech Republic

solis@dubito-ergo.com

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Scientific status of this paper. In October 2025, the CQFT framework hypothesized that the golden ratio $\varphi \approx 1.618$ arises as an infrared-attractive renormalization group (RG) fixed point of the nonlocal field theory, yielding anomalous dimension $\eta \approx 0.809$. Subsequent systematic simulation at $N = 200$–1000 produced a clean negative result: φ is not a preferred spectral scaling base. The φ-advantage lies in $[-0.29, -0.10]$ across all conditions — φ is consistently the *worst* spectral base, not the best.

This paper corrects that hypothesis. We show analytically that smooth RG flow *suppresses* multi-scale structure rather than generating it. We then demonstrate that φ arises from a qualitatively different mechanism — constraint-driven scale competition — which is consistent with all simulation results and provides a stronger theoretical foundation for the role of quasiperiodic geometry in the CQFT framework.

Abstract

We investigate the emergence of multi-scale structure in self-referential dynamical systems coupled to an evolving geometric substrate. Prior work hypothesized that renormalization group (RG) dynamics could generate the golden ratio φ as a nontrivial infrared fixed point. We show analytically that smooth RG coarse-graining instead suppresses scale hierarchy, converging to single-scale fixed points. Persistent multi-scale structure arises from a qualitatively different mechanism: constraint-driven dynamics in which competing scales cannot be simultaneously optimized. Using results from Diophantine approximation, KAM theory, and quasi-periodic spectral analysis, we show that φ emerges as the uniquely stable invariant under iterative scale competition, maximizing resistance to rational approximation.

We connect this result directly to the CQFT simulation findings: the Penrose quasicrystal, whose vertex set has φ-ratio geometry built in, achieves the lowest fractal embedding gap $\Gamma = |D_f - D_c|$ across all geometries tested ($N = 200$–1000), and is the only geometry supporting productive far-from-equilibrium dynamics. This is not coincidental: Penrose geometry instantiates the constraint-driven φ regime at the substrate level, providing the field with a natural multi-scale scaffold. We establish a clear separation: RG dynamics explain universality and scale elimination; constraint-driven dynamics explain persistent quasiperiodic structure. Five falsifiable predictions follow.

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1 Introduction

1.1 Two mechanisms for structure

Understanding the origin of structure in complex systems requires distinguishing two fundamentally different generating mechanisms:

- **Renormalization dynamics**, which reduce degrees of freedom by successive coarse-graining, producing universal scaling behaviour at fixed points.
- **Constraint-driven selection**, in which competing requirements prevent full optimization and enforce the coexistence of multiple scales.

Prior work in the CQFT programme ("Quantum Field Theory of Consciousness: φ Fixed-Point", 2025) proposed that φ arises via the first mechanism. The present paper demonstrates that φ arises via the second, and that this distinction has direct consequences for how the role of quasiperiodic geometry in self-referential field dynamics should be understood.

1.2 Why the correction strengthens the theory

The falsification of the RG fixed-point hypothesis is not merely a negative result to be buried. It is information: it tells us that φ's role in the CQFT framework is *not* that of a universal RG attractor (which would imply it should appear everywhere, under all conditions), but rather that of a *conditional structural invariant* (which appears only when the dynamics operate under competing scale constraints).

This is in fact consistent with all simulation findings:

1. φ is not a preferred spectral scaling base (negative): consistent with φ not being an RG fixed point.
2. Penrose geometry — which *does* have φ-scaling built into its vertex set — achieves the lowest Γ and the most productive dynamics: consistent with φ arising from constraint-driven selection at the substrate level.

The constraint-driven account is both more precise and more falsifiable than the RG account.

1.3 Structure of the paper

Section 2 establishes the field-theoretic setting. Section 3 shows why RG flow eliminates scale hierarchy. Section 4 develops the constraint-driven mechanism and derives φ as its invariant. Section 5 connects this to the CQFT simulation results via Penrose geometry. Section 6 states five falsifiable predictions. Section 8 draws conclusions.

2 Field-Theoretic Framework

We consider a scalar field $C(x,t)$ evolving on a graph $G = (V, E)$ with $|V| = N$ nodes, governed by:

$$\partial_t C = \mathcal{F}[C] + \eta(x,t) \tag{1}$$

where $\mathcal{F}[C]$ encodes local diffusion, nonlocal memory coupling, and negentropy-driven amplitude dynamics, and η represents stochastic perturbations with $1/f$ power spectrum.

The system is coupled to a geometric substrate characterized by a set of interaction scales $\{\alpha_i\}$. The full joint dynamics are captured by the AGI Lagrangian [1]:

$$\mathcal{L}_{\text{AGI}} = \lambda \left(-\sum_{i,j} W_{ij}(\mathbf{x}) \cos(\theta_i - \theta_j) \right) + (1-\lambda)\, E_{\text{sym}}(\mathbf{x}) \tag{2}$$

where $\lambda \in [0,1]$ governs phase-geometry coupling and E_{sym} is the quasiperiodic symmetry energy. The coupling term creates the scale competition that, as we show below, selects φ as the structural invariant.

3 Renormalization Group Dynamics and Scale Elimination

3.1 Wilsonian coarse-graining

Under Wilsonian renormalization, fast degrees of freedom above a momentum cutoff Λ are integrated out, producing an effective action at scale $\Lambda' < \Lambda$. Schematically:

$$e^{-S_{\text{eff}}[\phi_<]} = \int \mathcal{D}\phi_>\, e^{-S[\phi_< + \phi_>]} \tag{3}$$

This procedure generates RG flow equations for the coupling constants. In the nonlocal field theory with kernel $G(r) \sim |r|^{-\alpha}$, the flow of α under coarse-graining satisfies:

$$\frac{d\alpha}{d\ln\Lambda} = \beta(\alpha) \tag{4}$$

with $\beta(\alpha) < 0$ for $\alpha > d/2$ in spatial dimension d. This means the scaling exponent flows toward smaller values, which corresponds to *more local* effective interactions.

3.2 Scale hierarchy is not preserved

The key observation is that RG coarse-graining is designed to produce universality: it identifies which features of a system are irrelevant (flow to zero) and which are relevant (flow to fixed points). This is precisely what makes RG powerful for critical phenomena.

But this same property makes RG unsuited to generating multi-scale structure. If a system has two competing scales α_L and α_S with $\alpha_L > \alpha_S$, RG flow treats the smaller scale as a subleading correction and drives the effective theory toward a single dominant scale:

$$\text{RG}: \quad (\alpha_L, \alpha_S) \longrightarrow (\alpha^*, 0) \tag{5}$$

Numerical investigation confirms this: initializing with two distinct scales and running the RG flow, the ratio $r = \alpha_L/\alpha_S$ does not stabilize at φ but instead diverges as α_S flows toward zero. We record this as:

Result 1 (RG scale collapse). Smooth RG coarse-graining of the CQFT nonlocal field theory eliminates scale hierarchy. No stable irrational ratio, including φ, emerges under smooth RG flow. Therefore:

$$\text{RG dynamics} \Rightarrow \text{single-scale fixed point.}$$

The hypothesis that φ is an infrared-attractive RG fixed point is **falsified** by this analysis and by systematic simulation.

4 Constraint-Driven Multi-Scale Selection

4.1 Competing constraints prevent scale collapse

We now introduce the regime that RG analysis misses: systems in which structural constraints prevent full optimization of either scale independently. In the CQFT context, these constraints arise naturally from the AGI Lagrangian (2): the phase-coupled drift term drives the system toward phase synchronization (a single-scale coherent state), while the symmetry relaxation term drives it toward quasiperiodic order (which requires multi-scale structure). Neither term can dominate completely at $\lambda > 0$; they compete.

Formally, define two scales $\alpha_L > \alpha_S > 0$ and constraints:

$$\alpha_L \text{ cannot collapse to } \alpha_S \quad \text{(phase coherence prevents this)} \tag{6}$$

$$\alpha_S \text{ cannot expand to } \alpha_L \quad \text{(symmetry relaxation prevents this)} \tag{7}$$

The system must find a stable ratio $r = \alpha_L/\alpha_S$ that satisfies both constraints simultaneously.

4.2 Commensurability collapse and its avoidance

A ratio is *commensurate* if:

$$r \approx \frac{p}{q}, \quad p, q \in \mathbb{Z}^+ \tag{8}$$

Commensurate ratios produce resonance: the two scales lock, reducing the effective number of independent degrees of freedom and collapsing the multi-scale structure. This is the discrete analogue of the single- scale RG fixed point.

The constraint system must therefore avoid commensurability. Following Diophantine approximation theory, we impose a lower bound on the separation from any rational:

$$\left| r - \frac{p}{q} \right| \geq \frac{c}{q^2} \quad \forall\, p/q \in \mathbb{Q} \tag{9}$$

Numbers satisfying (9) are called *badly approximable* or Diophantine. The question is: which badly approximable number is the *most* stable under the constraint dynamics?

4.3 The golden ratio as the optimal invariant

Definition 4.1 (Approximation resistance functional)**.**

$$\mathcal{F}(r) = \inf_{p/q \in \mathbb{Q}} \left(q^2 \left| r - \frac{p}{q} \right| \right) \tag{10}$$

This measures the worst-case resistance to rational approximation of the number r.

Theorem 4.2 (Hurwitz; see 2)**.** *For any irrational r:*

$$\mathcal{F}(r) \leq \frac{1}{\sqrt{5}} \tag{11}$$

with equality if and only if $r = \varphi$ (or its algebraic conjugate). Therefore:

$$r^* = \arg\max_{r \notin \mathbb{Q}} \mathcal{F}(r) = \varphi \tag{12}$$

φ is the hardest *irrational to approximate by rationals, and is thus maximally resistant to commensurability collapse.*

Result 2 (Constraint-driven φ selection). In a system subject to competing scale constraints (6)–(7) with commensurability avoidance (9), the unique stable scale ratio is:

$$r^* = \varphi = \text{ratio maximizing resistance to rational approximation.}$$

This is *not* an RG fixed point. It is a structural invariant of constraint-driven scale competition.

4.4 Dynamical characterization: iterative scale mixing

The constraint dynamics can be modeled as iterative scale mixing. When the two constraints interact recursively, the scale ratio evolves as:

$$r_{n+1} = 1 + \frac{1}{r_n} \tag{13}$$

This is the continued fraction recursion. Its unique fixed point is:

$$r^* = 1 + \frac{1}{r^*} \implies r^* = \varphi \tag{14}$$

This provides a dynamical derivation of φ as the *only* ratio stable under iterative scale competition, independent of the Diophantine argument.

4.5 Extensions: KAM theory and quasi-periodic spectra

Circle maps and KAM. In the driven oscillator system:

$$\theta_{n+1} = \theta_n + \Omega + K \sin(2\pi\theta_n) \tag{15}$$

the last invariant torus to survive as the coupling K increases occurs at the winding number $\Omega = \varphi^{-1}$. By the KAM theorem [3, 4, 5], this torus persists longest because φ^{-1} is the most badly approximable rotation number. Its destruction marks the onset of global chaos. This is the continuous-time analogue of the constraint-driven selection.

Quasi-periodic spectra. For potentials of the form:

$$V(x) = V_0 \cos(2\pi x) + V_1 \cos(2\pi r\, x) \tag{16}$$

irrational r produces quasi-periodic (Cantor-like) spectra. Spectral stability is maximized when r is maximally irrational. Penrose tilings, whose vertex sets have ratio φ between the two rhombus edge lengths, realize this condition geometrically: they produce the characteristic 5-fold / 10-fold diffraction peaks of maximally stable quasi-periodic order.

5 Connection to CQFT Simulation Results

5.1 Penrose geometry as physical instantiation

The Penrose quasicrystal is not just a mathematical example: it is the geometry that instantiates φ-ratio scale competition at the substrate level. A Penrose tiling has exactly two rhombus types with edge ratio φ. Its vertex set has fractal scaling consistent with φ (the amplitude-weighted correlation dimension $D_c^{\text{Penrose}} \in [1.41, 1.49]$ across $N = 200$–1000, near $\sqrt{2} \approx 1.414$, the logarithmic mean of 1 and 2 under φ-weighting).

The constraint-driven account now explains the key simulation finding:

Why Penrose achieves the lowest Γ. The embedding gap $\Gamma = |D_f - D_c|$ measures the mismatch between the field's external and internal descriptions of its spatial structure. On a Penrose substrate, the field operates in the constraint-driven φ regime: the two competing scales (phase coherence drives toward synchronization; symmetry relaxation drives toward quasiperiodic order) are balanced at the maximally stable ratio. This allows the field to maintain *both* spatial complexity (D_f) and coherent self-correlation (D_c) simultaneously, minimizing their mismatch.

On a square lattice, there is no such balance: the 4-fold symmetry imposes a commensurate structure that frustrates the field's natural quasiperiodic tendency, producing large Γ.

5.2 Reconciliation with the φ negative result

The simulation found that φ is not a preferred spectral scaling base. This is fully consistent with the constraint-driven account:

- The spectral scaling test asks whether φ appears in the *field's temporal spectrum* as an RG fixed point would. It does not — because φ is not an RG fixed point.
- The constraint-driven account predicts φ in the *substrate's geometric ratio* — which the Penrose geometry does instantiate, and which does produce the lowest Γ.

The two results are not only consistent but mutually confirming: the negative result rules out the RG mechanism; the positive Penrose result confirms the constraint-driven mechanism operating at the geometric level rather than the spectral level.

5.3 Geometry-ordering result

Table 1 summarizes the three dynamical regimes in terms of their constraint structure.

Table 1: Three substrate geometries, their constraint structure, and simulation observables ($N = 500$, $\beta = 4.2$).

	Random	**Square lattice**	**Penrose**
Scale structure	Unconstrained (no competing scales)	Commensurate (4-fold frustrated)	Constraint-driven φ
Γ	0.19 (moderate)	0.87 (large)	**0.20 (stable, small)**
Prigogine regime	Near-equilibrium	Frustrated far-from-equil.	**Productive far-from-equil.**
$\mathcal{D}$ (dubito)	≈ 0 (absorbed)	0.09 (overworking)	**0.08 (efficient)**
D_c stability	Monotone growth	Non-monotone	**Stable [1.41, 1.49]**

6 Falsifiable Predictions

The constraint-driven account generates the following falsifiable predictions, distinguishable from those of the (falsified) RG account:

P1. Spectral base test (already confirmed): φ is not a preferred temporal spectral scaling base in the CQFT field equations. This follows from the absence of a φ RG fixed point. *Status: confirmed experimentally; φ-advantage $\in [-0.29, -0.10]$.*

P2. Geometric ratio test: Substrates whose vertex sets embed a φ edge ratio (Penrose and 3D icosahedral quasicrystals) should achieve systematically lower Γ than substrates with rational or unconstrained geometry, *at the same N and β. Status: confirmed; $\Gamma_{lattice} \gg \Gamma_{Penrose} \approx \Gamma_{random}$ at every N tested.*

P3. KAM analogue test: In a simulation of two coupled CQFT oscillators with variable frequency ratio r, the last stable quasi-periodic orbit before resonance-driven synchronization should occur near $r = \varphi^{-1}$. This would be the direct CQFT analogue of the KAM circle-map result. *Status: not yet tested; proposed as next experiment.*

P4. Disruption test: Perturbing a Penrose substrate toward a nearby rational approximant (e.g., the Fibonacci lattice at ratio $F_{n+1}/F_n \to \varphi$) should increase Γ monotonically as the substrate approaches commensurate structure, with the maximum Γ at the exact rational. *Status: not yet tested.*

P5. AGI applicability: In transformer models with RoPE or ALiBi positional encoding — which introduce a geometric length-scale dependence into the attention weights — the layer-wise joint energy gradient $\delta\mathcal{L}^{(l)} = \mathcal{L}^{(l)} - \mathcal{L}^{(l-1)}$ should be more negative than in models with absolute positional embeddings, because RoPE/ALiBi instantiate a partial constraint-driven scale competition via the rotational frequency ratio. The natural frequency ratio of optimal RoPE encoding, under this account, would approach φ in the limit of maximum representational efficiency. *Status: predicted; testable on existing open-weight models without modification.*

7 Theoretical Consolidation

7.1 The separation principle

The two mechanisms are now clearly separated:

Separation Principle.

$$\begin{aligned} \text{RG dynamics} &\Rightarrow \text{scale elimination}, \quad \varphi \text{ does not arise.} \\ \text{Constraint-driven dynamics} &\Rightarrow \text{scale coexistence}, \quad \varphi = r^* \text{ is the unique stable invariant.} \end{aligned}$$

φ is not a universal constant of dynamics. It is a conditional structural invariant: it appears if and only if the system operates under competing scale constraints with commensurability avoidance. Penrose and icosahedral quasicrystals are physical instantiations of this condition.

7.2 Relation to the ouroboros conjecture

The constraint-driven φ result provides a geometric interpretation of the ouroboros conjecture (that a self-referential field selects its own consistent substrate). The field on a Penrose substrate is already operating in the constraint-driven regime: the phase-coupled drift and symmetry relaxation compete at the φ ratio. The field does not need to "search" for this geometry through neutral evolution (as the falsified strong form of the conjecture assumed); it needs a substrate that already embeds the constraint structure. Penrose geometry provides this. This is why the conditional form of the ouroboros conjecture is confirmed (phase-gradient coupling + Penrose substrate $\rightarrow$ self-organization) while the strong form is falsified (neutral evolution alone is insufficient).

7.3 Relation to the geometric FEP

The operational geometric free energy $\mathcal{F}_{\text{geo}} = \Gamma$ is minimized on Penrose geometry. In the constraint-driven account, this is because Penrose geometry minimizes the mismatch between D_f and D_c by providing the field with a substrate whose natural scaling ratio (i.e., φ) matches the ratio that minimizes commensurability collapse. The geometric FEP is thus grounded in the Diophantine optimality of φ, not in its appearance as an RG fixed point.

8 Conclusions

We have established the following:

1. Smooth RG flow suppresses multi-scale structure and does not generate φ as a fixed point.
2. φ arises from constraint-driven scale competition as the unique maximally stable invariant under commensurability avoidance.
3. This is consistent with all CQFT simulation results: the negative spectral base result rules out the RG mechanism; the Penrose geometry result confirms the constraint-driven mechanism.
4. The constraint-driven account provides a stronger and more falsifiable foundation for the role of quasiperiodic geometry in self-referential field dynamics than the original RG hypothesis.

The general conclusion is:

$$\text{RG} \rightarrow \text{scale elimination}, \qquad \text{constraints} \rightarrow \text{scale coexistence}. \tag{17}$$

$$\varphi = \text{optimal invariant preserving multi-scale structure under competing constraints.} \tag{18}$$

The Principled Field Theory of Consciousness (PFT) and its core mathematical structure are unaffected by this correction. The correction applies specifically to the claim that φ is an infrared RG fixed point. The revised account is both mathematically sounder and more directly connected to the empirical results.

Acknowledgments

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